MIL-I-19105(Aer)
23 September 1955
Superseding
NAVORD OS 3855
9 August 1945

MILITARY SPECIFICATION

INTERVAL GENERATOR MARK 1 MOD 0

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification establishes the requirements for the manufacture, inspection and test procedure of a timing unit designed to generate and supply timed electrical impulses to a one wire bomb release system, which is used with and controlled by the Interval Selector Mark 1 and Mods, and shall be known as Interval Generator Mark 1 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 The following specification, standards, drawings and publications, of the issue in effect on the date of invitation for bids, forms a part of this specification.

SPECIFICATIONS:

Federal

QQ-M-151	Metals: General Specification for Inspection of

Military

MIL-D-5028 (ASG)	Drawings and Data Lists; Preparation of Manufacturer's (For Production Aircraft, Guided Missiles, Engines, Accessories and other Auxiliary Equipment)
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-	Interval Generator Mark 1 Mod 0

Navy Department

General Specification for the Inspection of Materials

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DRAWINGS:

P. R. Mallory and Company Drawing No. C160095	Impulse Generator, General Arrangement and all Drawings and Specifications Listed Thereon

PUBLICATIONS:

Air Force Navy Aeronautical Bulletin No. 143	Specifications and Standards; Use of

(When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention, Technical Records Division.)

3. REQUIREMENTS

3.1 Materials: Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.1.1 Selection of Materials: Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143.

3.1.1.2 Standard Parts: Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.1.2 Critical and Strategic Materials: Materials shall be chosen on the basis of suitability and availability in this country in quantity and from consideration of the additional restrictions created during a national emergency. Critical and strategic materials shall be used only to the extent considered mandatory for satisfactory operation. The contemplated use of these materials shall be brought to the attention of the procuring service with appropriate justification therefor and shall be subject to approval of the procuring service.

3.2 Conformance to Drawings: No departure from revision or change to the drawings specified herein shall be permitted without prior written approval by the Procuring Service. In the event changes are approved, (see paragraph 6.1) the contractor shall furnish the Procuring Service with reproducible drawings covering the change. Conformance to drawings will not relieve the contractor of the requirements of paragraphs 3.3 and 3.5 below.

3.3 Manufacturing and Finish: Manufacturing processes and finishes shall be as shown on the applicable drawings unless otherwise approved. (See paragraph 6.1). The responsibility shall be assumed by the contractor for any malfunction or failure of the component parts or the assembled equipment to pass the required tests due to unsatisfactory fits, clearance, tolerances, internal stresses, or finishes.

3.3.1 Weight Reduction: The contractor shall make every effort to reduce the over-all weight without impairing the equipment's function, operation or durability. All changes that are required to accomplish weight reduction shall be subject to written approval by the Procuring Service prior to incorporation.

3.4 Interchangeability: All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

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3.5 Workmanship: All details of workmanship shall be in accordance with highgrade armament manufacturing practice for the quantity production of this type of equipment. All parts shall be free from excessive marks, grooving, indentations, cracks, flaws, and any other injurious defects which may impair the serviceability of the assembled equipment. All workmanship shall be of a quality satisfactory to the Procuring Service. Where it is necessary to include additional data, descriptive and appropriate headings will be used and arranged in a logical sequence.

3.6 General Operating Conditions: The Interval Generator shall be capable of functioning properly under any or all combinations of the following operating conditions:

3.6.1 Altitude: From sea level to 50,000 feet.

3.6.2 Temperature: From -75° to +125°F.

3.6.3 Humidity: Under all conditions of relative humidity at temperatures found from sea level to 50,000 feet.

3.6.4 Acceleration, Vibration and Shock: Under all conditions of accelerations, vibration and shock that are present in aircraft during service operations.

3.6.5 Altitude: Tilted or oriented in any manner.

3.7 Performance: All performance requirements listed in this specification are for the Interval Generator when used with and controlled by an Interval Selector Mark 1 Mod 0 whose resistors are within +1% of the nominal values listed in paragraph 3.7.2 of MIL-I-_____(Aer).

3.7.1 Voltage: The Interval Generator shall operate when energized at its terminals by a 22 to 28-1/2 volt direct current under all conditions specified herein.

3.7.2 Frequency Range: The Interval Generator shall be capable of producing a range of impulses from 2 per second to 20 per second with nominal values in accordance with the chart shown below:

Switch Position	Pulses Per Second
1	2.000
2	2.244
3	2.618
4	2.826
5	3.17
6	3.556
7	3.99
8	4.478
9	5.024
10	5.636
11	6.325
12	7.096
13	7.962
14	8.934
15	10.024
16	11.246
17	12.620
18	14.159
19	15.896
20	17.825
21	20.000

3.7.3 Frequency Accuracy: The frequency of the Interval Generator shall have an error not exceeding ± 25% with an applied emf of 24 to 28-½ volts direct current in an ambient temperature range of -70°F to -125°F with an applied emf of 26 volts direct current, and in an ambient temperature of 75°F ± 10°F the maximum error shall not exceed ± 10% in any impulse setting of the Interval Selector switch.

3.7.4 Current: The total average current consumption of the unit, including that drawn by the heating element, shall not exceed 4 amperes at 28 volts direct current.

3.7.5 On-Off Time: When the unit is operated at 26 volts D. C. and at 75° ± 10°F, the "on time" or "off time" shall be not less than .02 seconds for any position of the Interval Selector Switch Mark 1 Mod 0.

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3.7.6 Initial Lag: The time that elapses from the closure of the Bomb Release Switch and the initiation of the first impulse shall be not less than .015 second when measured at 26 volts D. C. and at 75°F ± 10°F.

3.7.7 First and Last Impulses: The Impulse Generator shall be capable of operating in such a manner that the first and last impulse is not cut short of its intended duration.

3.7.8 Connections of Parallel Electric Circuits: The Impulse Generator shall be constructed so that the connection of parallel circuits drawing 15 amperes or less to any of the connector pins, except those in the timing circuit, either accidently or purposely, shall have no detrimental effect upon the operation of the unit or upon any of its parts as long as the emf applied to the timing circuits remains between 24 to 28½ volts direct current.

3.7.9 Output Capacitor: The output contacts may be shunted by a capacitor, not to exceed one microfarod in capacity, in order to reduce arcing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests: The inspection and testing of the Interval Generator shall be classified as follows:

a. Preproduction Tests: Preproduction Tests are those tests accomplished on samples which are representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification.

b. Inspection Tests: Inspection Tests are those tests accomplished on the Interval Generator manufactured and submitted for acceptance under contract.

4.2 Preproduction Tests:

4.2.1 Sampling Instructions: The first four (4) Interval Generators under each contract shall be preproduction test samples and shall be delivered for inspection and testing as directed, together with a certification by the cognizant inspector that the materials used in their manufacture are in accordance with pertinent specifications.

4.2.2 Tests: The preproduction samples shall be checked for strict conformance with the pertinent drawings and specifications, and shall be subjected to any or all tests of this specification, as described under "Test Methods."

4.2.3 Correction of Defects: Any defects discovered during the inspection and testing of the preproduction samples shall be corrected by the contractor. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent Interval Generators offered for acceptance under the contract.

4.3 Inspection Tests: The equipment, together with all the materials used in the manufacture thereof, shall be subjected to inspection at the contractor's plant, or elsewhere, by authorized Government inspectors. Inspection tests shall consist of Individual and Sampling tests.

4.3.1 Materials Inspection:

4.3.1.1 Commercial Materials: When materials are procured under commercially recognized non-Government specifications, such as by SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of such material. Such certificate shall be in a form acceptable to the Government Inspector.

4.3.1.2 Approval of Materials in Process: Inspection and approval of material in process during the course of manufacture shall not be construed as insuring the acceptance or approval of the finished product.

4.3.2 Individual Tests: Each Interval Generator shall be subjected to the following tests, as described under "Test Methods." These Individual tests shall be performed by the contractor.

a. Examination of product

b. Functional tests

4.3.3 Sampling Tests: The following number of production units shall be selected at random

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from each production lot and subjected to the tests listed below, as described under "Test Methods":

No. of Interval Generators in Lot	No. of Sample Interval Generators
99 or less	2
100 to 299	3
300 to 500	4

4.3.3.1 Shock Test: Two (2) preproduction samples in turn shall be secured, in each of their six (6) rectangular positions in turn, to the carriage of an Impact Drop Tester, similar to the type used by the Naval Ordnance Laboratory, United States Naval Gun Factory, Washington 25, D. C. The carriage shall be dropped through such height that when decelerated by impact, a deceleration of a minimum of 200 gravitational units is recorded on an Accelerometer. This Accelerometer shall be similar to the Mine Unit, Accelerometer Mark 1 of the Naval Ordnance Laboratory. When the unit is connected to 24 to 28.5 volts D. C., the closure of the outlet relay during the test, or any other electrical or mechanical malfunctioning or derangement, shall be considered an indication of failure.

4.3.3.2 Corrosion Resistance: Two (2) preproduction samples (not the two tested in paragraph 4.3.3.1) shall be subjected to a salt spray test as specified in Specification QQ-M-151 for 120 hours. The unit shall function properly and shall show no signs of excessive corrosion or leakage at the termination of this test.

NOTE: The electrical receptacle shall be blocked off or capped by some suitable means during this test.

4.3.4 Rejections:

4.3.4.1 Sampling Test Samples: The failure of any Sampling test samples to pass the inspection and tests outlined in paragraph 4.3.2 shall cause tentative rejection of all Interval Generators represented by the sample, subject to retest as hereinafter provided.

4.3.4.2 Individual Test Units: The failure of any Interval Generator, any piece, subassembly, or assembly of an Interval Generator to pass the inspection and tests outlined in paragraphs 4.5.1 thru 4.5.4.2 shall cause its tentative rejection subject to retest as hereinafter provided.

4.3.5 Retests:

4.3.5.1. Sampling Test Samples - First Retest: In case any sample representing a production lot fails to pass the tests outlined in paragraph 4.3.3, the contractor shall have the opportunity of reworking the entire lot affected and resubmitting to retest twice as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.3.5.2 Samrling Test Samples - Second Retest: If, in the inspectors opinion, the failure of the retest samples is due to a remediable defect, the contractor may, at the discretion of the Inspector, be offered the opportunity of reworking the lot affected and resubmitting to a second retest three times as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.3.5.3 Individual Test Units: Any Interval Generator or any piece, subassembly, or assembly of an Interval Generator which fails to pass the tests as outlined in paragraphs 4.5.1 thru 4.5.4.2 may, upon the request of the contractor, be resubmitted for individual inspection and test, provided a statement showing the corrective measures taken is furnished to the inspector at the time of such resubmission. Upon retest, failure to meet the requirements outlined in paragraphs 4.5.1 thru 4.5.4.2 shall be cause for final rejection of the lot represented.

4.3.5.4 Salvaged Pieces: Any piece from a rejected Interval Generator may be salvaged for use provided each salvaged piece is inspected for, and is in complete compliance with, the pertinent drawings and specifications listed on P. R. Mallory Dwg. C160095.

4.4 Inspection Subsequent to Delivery: Any or all Interval Generators, subsequent to the first preproduction samples, may be subject to inspection and testing for conformance with all requirements and tests of this specification after delivery to the Government.

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4.5 Test Methods:

4.5.1 Examination of Product: Each Interval Generator offered for acceptance under the contract shall be inspected to insure that it has been properly assembled and adjusted, and that the workmanship and finish are in keeping with the intent of this specification and applicable drawings in accordance with the requirements of paragraph 3.5. In addition each and every Interval Generator shall be subject to, and shall satisfactorily pass the following tests conducted at an ambient temperature of 75° ± 10°F.

4.5.1.1 Accuracy and Lag Test: Each unit shall be tested at positions 1, 7, 14 and 21 of the Interval Selector Mark 1 Mod 0, for frequency accuracy and initial lag at 26 volts D. C. as specified in paragraphs 3.7.3 and 3.7.4 respectively.

4.5.1.2 "ON-OFF" Time Test: Each unit shall be tested, at positions 15 and 21 of the Interval Selector Mark 1 Mod 0, for "on-off" time at 26 volts D. C., as specified in paragraph 3.7.5.

4.5.1.3 Dielectric Breakdown Test: Each unit shall have all ground connections to the case broken after which all of the pins of the electrical connector receptacle shall be connected to a common point which is not connected to "ground". An alternating test voltage of 500 volts (rms) shall then be applied between this common point and "ground". The voltage shall be maintained at 500 volts (rms) for a period of one minute. The frequency shall be any value between 50 and 100 cycles per second, and the shape of the wave shall be such as conforms to good commercial practice. The voltmeter method shall be used to measure the test voltage. The voltmeter shall derive its potential either directly from the high voltage circuit through an auxiliary transformer, or by means of the voltmeter coil being placed into the primary circuit of the testing transformer. In lieu of the above, it is permissible to use a standard commercial dielectric testing transformer subject to approval of the Bureau of Aeronautics. Indication of the flow of current as evidenced by some suitable indicating device, shall be indicative of failure.

4.5.1.4 Minimum Voltage Operation: Each unit shall be tested for operation at all positions of the Interval Selector when energized with an applied emf of 22 volts.

4.5.2 Insulation Resistance: The Interval Generator shall be subjected to an ambient temperature of 125°F for a period of one hour. The insulation resistance shall be measured while the unit is at this temperature by means of a constant potential "megger", under a potential of 500 volts direct current, or other suitable method, applied between all of the receptacle pins connected in parallel and the case (all ground leads to the case shall be broken for this test). The unit shall then be removed from the heat, and allowed to cool in a 90% moist atmosphere until stabilized at room temperature, after which the insulation resistance shall be measured as described above. Under either temperature condition, a measured insulation resistance of less than one megohm shall be cause for rejection of the unit.

4.5.3 Operational Test: The Interval Generator shall be energized with an emf of 28½ volts direct current and set, by means of an Interval Selector, Mark 1 Mod 0, to deliver impulses at a rate of 20 cycles per second. The unit shall then be given an operational test as follows:

4.5.3.1 2500 impulses in runs of 100 impulses with a 15 ampere inductive load equivalent to three Navy Mark I release solenoids connected in parallel at atmosphere pressure and at room temperature with a 20 second cooling period between runs.

4.5.3.2 4500 impulses with a 5 ampere inductive load equivalent to one Navy Mark I release solenoid under the same conditions.

4.5.3.3 2500 impulses in runs of 100 impulses with a 15 ampere inductive load equivalent to three Navy Mark I release solenoids connected in parallel at atmosphere pressure and at room temperature with a 20 second cooling period between runs.

4.5.4 Vibration Tests:

4.5.4.1 Natural Frequency Vibration Test: The Interval Generator shall be connected to a 24 to 28.5 volt D. C. source of power, placed on a vibration stand and in turn vibrated in each of its three principal axes to determine excessive occurence of any natural frequency of vibration of component parts. Vibration shall follow approximately a simple harmonic motion and the frequency shall be gradually increased from 600 cycles per minute to 3300 c.p.m. and return to 600 c.p.m. within a period of not less than one minute. The vibration shall have an amplitude of .003 inch (.006 inch total excursion or displacement).

During four successive periods of vibration, the unit shall be operated in turn at

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positions 1, 7, 14, and 21 of the Interval Selector Mark 1 Mod 0 and a record taken of the output pulses. The frequency and On-Off time shall conform to the requirements of paragraphs 3.7.3 and 3.7.5.

4.5.4.2 Destructive Vibration Test: After the test specified in paragraph 4.5.4.1, the unit shall be vibrated in each of its three principal axes, in turn for a period of nine hours (three hours in each axis). The vibration shall follow approximately a simple harmonic motion and the frequency shall be maintained at 3000 cycles per minute. The vibration shall have an amplitude of .015 inch (.030 inch total excursion or displacement). Any electrical or mechanical failure or loosening of parts shall be considered as a failure of the unit to pass the test. At the completion of this test, the unit shall be tested at positions 1, 7, 14, and 21 of the Interval Selector Mark 1 Mod 0 for timing accuracy and On-Off time as specified in paragraphs 3.7.3 and 3.7.5.

5. PREPARATION FOR DELIVERY

5.1 Application: The preservation, packaging and marking requirements specified herein apply only to direct purchases by or shipments to the Government.

5.2 Preservation, Packaging, Packing and Marking: Each Interval Generator shall be preserved, packaged, packed and marked in accordance with Group X, Type 2 of Specification MIL-P-6064.

6. NOTES

6.1 Intended Use:

6.2 Adherence to Drawings and Specifications - Conflict Elimination: Insofar as the contractor is required by provisions of this specification to adhere rigidly to the drawings listed in Section 2, which may make it impossible in some instances also to comply with the performance and test requirements specified herein, then, in that event, the contractor should promptly notify the procuring activity of such fact, and appropriate changes will be made in the applicable drawings, parts, or performance requirements as authorized by the procuring activity. An equitable adjustment, in accordance with applicable contract provisions, will be made in the price of the articles to be delivered to reflect any change so authorized by the procuring activity.

6.3 Ordering Data: Each bid submitted in connection with the procurement of the Interval Generators shall be accompanied by a statement showing whether or not any exceptions are being taken to the requirements shown in the drawings or in this specification. If exceptions are taken, they shall be specifically described, and the alternative thereto indicated in the statement.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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